Ch.3 Terminology

Ch.3 Terminology:

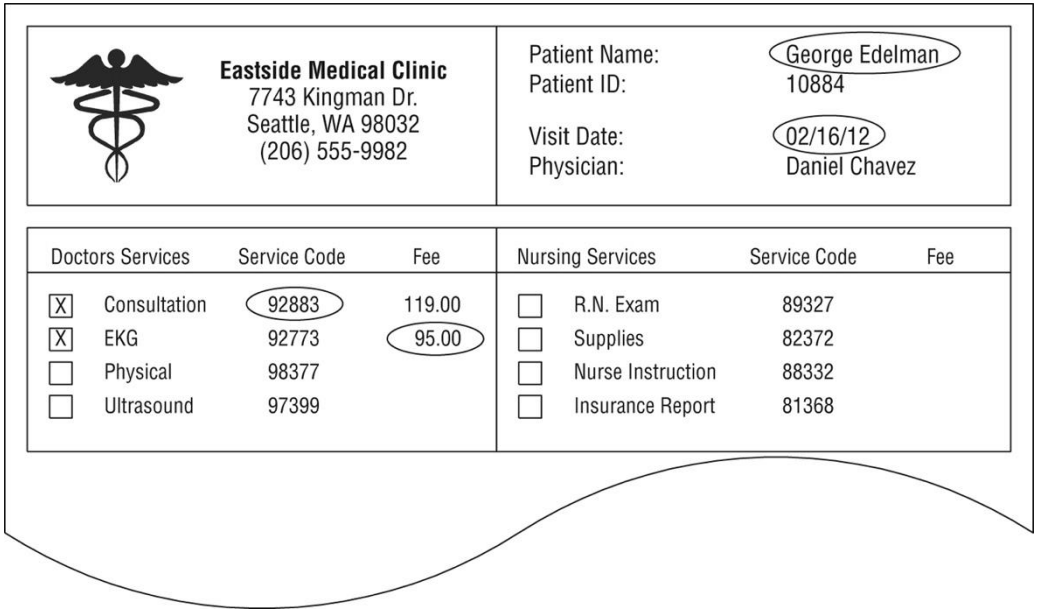
A. Value-Related Terms
B. Structure-Related Terms
C. Relationship-Related Terms
D. Integrity-Related Terms

A. Value-Related Terms

1. Data vs Information (processed data)
   - Data is what you store; information is what you retrieve.
2. Null – missing or unknown values

```
George Edleman     92883     05/16/96     95.00
```

Figure 3.1 *An example of basic data*



Eastside Medical Clinic
7743 Kingman Dr.
Seattle, WA 98032
(206) 555-9982

Patient Name: George Edelman
Patient ID: 10884
Visit Date: 02/16/12
Physician: Daniel Chavez

Doctors Services	Service Code	Fee	Nursing Services	Service Code	Fee
[X] Consultation	92883	119.00	[ ] R.N. Exam	89327	
[X] EKG	92773	95.00	[ ] Supplies	82372	
[ ] Physical	98377		[ ] Nurse Instruction	88332	
[ ] Ultrasound	97399		[ ] Insurance Report	81368	

Figure 3.2 *An example of data transformed into information*

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Anomalies created by null

Products

Product ID	Product Description	Category	SRP	Qty On Hand	Total Value
70001	Shur-Lok U-Lock	Accessories	75.00		
70002	SpeedRite Cyclecomputer		65.00	20	1,300.00
70003	SteelHead Microshell Helmet	Accessories	36.00	33	1,118.00
70004	SureStop 133-MB Brakes	Components	23.50	16	376.00
70005	Diablo ATM Mountain Bike	Bikes	1,200.00		
70006	UltraVision Helmet Mount Mirrors		7.45	10	74.50

Figure 3.4 *The nulls in this table will have an effect on mathematical operations involving the table's fields.*
[TOTAL VALUE] = [SRP] × [QTY ON HAND].

Category Summary

Category	Total Occurrences
	0
Accessories	2
Bikes	1
Components	1

Figure 3.5 *Nulls affect the results of an aggregate function.*

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B. Structure-Related Terms

1. Table

- Must include at least the primary key

Patient Visit

Patient ID	Visit Date	Visit Time	Physician	Blood Pressure	<< other fields >>
92001	02/14/12	10:30	Hernandez	120/80	.....
97002	02/14/12	13:00	Black	112/74	.....
99014	02/15/12	09:30	Rolson	120/80	.....
96105	02/15/12	11:00	Hernandez	160/90	.....
96203	02/15/12	14:00	Hernandez	110/75	.....
98003	02/16/12	09:30	Rolson	120/80	.....

Figure 3.7 *A table representing an event*

Clients

Client ID	Client First Name	Client Last Name	Client City	<< other fields >>
9001	Stewart	Jameson	Seattle	.....
9002	Susan	Black	Poulsbo	.....
9003	Estela	Rosales	Tacoma	.....
9004	Timothy	Ennis	Seattle	.....
9005	Marvin	Russo	Bellingham	.....
9006	Kira	Bently	Tacoma	.....

Records

Fields

Figure 3.6 *A typical table structure*

Categories

Category ID	Category Name
10000	Accessories
20000	Bikes
30000	Clothing
40000	Components

Figure 3.8 *An example of a validation table [it is rarely modified]*

2. Field (attribute)

Calculated Field ↓ Multipart Field ↓ Multivalued Field ↓

Clients

Client ID	Client First Name	Client Last Name	Client Full Name	Address	Client City, State, Zip	Account Rep
9001	Stewart	Jameson	Stewart Jameson	.....	Seattle, WA 98125	John, Sandi
9002	Susan	Black	Susan Black	.....	Poulsbo, WA 98370	Frits
9003	Estela	Rosales	Estela Rosales	.....	Bellevue, WA 98005	John
9004	Timothy	Ennis	Timothy Ennis	.....	Seattle, WA 98115	Frits, Sandi
9005	Marvin	Russo	Marvin Russo	.....	Bellingham, WA 98225	Frits, John
9006	Kira	Bently	Kira Bently	.....	Olympia, WA 98504	Sandi

Figure 3.9 *A table containing regular, calculated, multipart, and multivalued fields*
[last three types exist in poorly designed databases]

3. Record (tuple) = a row, ex. Susan Black record above

4. View = virtual table built of rows from one or more base tables

Students

Student ID	Student First Name	Student Last Name	Student Phone	<< other fields >>
60001	Zachary	Erlich	553-3992	.....
60002	Susan	Black	790-3992	.....
60003	**Joe**	**Rosales**	551-4993	.....

Student Instruments

Student ID	Instrument ID	Checkout Date
60002	1003	02/02/12
60001	1002	02/06/12
60003	1000	**02/06/12**

Instruments

Instrument ID	Instrument Description	Category	<< other fields >>
1000	**Stratocaster**	Guitar	.....
1001	Player 2100 Multieffects	Multieffect Unit	.....
1002	JCM 2000 Tube Super Lead	Amplifier	.....
1003	Twin Reverb Reissue	Amplifier	.....

Instrument Assignments (*View*)

Student First Name	Student Last Name	Instrument Description	Checkout Date
Zachary	Erlich	JCM 2000 Tube Super Lead	02/06/12
Susan	Black	Twin Reverb Reissue	02/03/12
Joe	**Rosales**	**Stratocaster**	**02/06/12**

Figure 3.10 *An example of a typical view*

5. Keys

- Primary key:

a field or a group of fields that uniquely identifies each record within a table

  – Every table should have one

- Foreign key:

a field that establishes relationships between two tables and it is the primary key in the other table

- [Foreign key can also help with enforcing constrains]

Agents

Primary Key

Agent ID	Agent First Name	Agent Last Name	Date of Hire	Agent Home Phone
100	Stella	Rosales	05/16/09	553-3992
101	Steve	Horst	10/15/09	790-3992
102	Samantha	Nathanson	03/01/11	551-4993

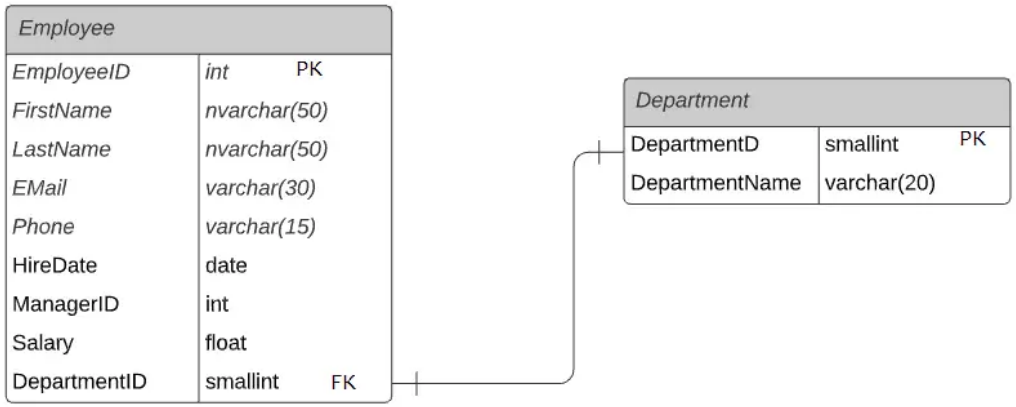
Foreign Key

Entertainers

Primary Key

Entertainer ID	Agent ID	Entertainer Name	Entertainer Phone	<< other fields >>
9001	100	Jazz Time	555-9928	.....
9002	101	The Mike Hernandez Trio	099-8837	.....
9003	100	The Country Squires	709-3542	.....

Figure 3.11 *An example of primary and foreign key fields*



Ex. You cannot add in table Employee the employee Anna form Physics dept., if Physics is not created in table Departments

C. Relationship-Related Terms

1. Relationships

One-to-One Relationships

Employees

Employee ID	Employee First Name	Employee Last Name	Home Phone	*<< other fields >>*
100	Zachary	Erlich	553-3992	.....
101	Susan	Black	790-3992	.....
102	Joe	Rosales	551-4993	.....

Compensation

Employee ID	Hourly Rate	Commission Rate	*<< other fields >>*
100	19.75	3.5%	.....
101	25.00	5.0%	.....
102	22.50	5.0%	.....

Figure 3.13 *An example of a one-to-one relationship*

One-to-Many Relationships

Agents

Agent ID	Agent First Name	Agent Last Name	Date of Hire	Agent Home Phone
100	Stella	Rosales	05/16/09	553-3992
101	Steve	Horst	10/15/09	790-3992
102	Samantha	Nathanson	03/01/11	551-4993

Entertainers

Entertainer ID	Agent ID	Entertainer Name	Entertainer Phone	*<< other fields >>*
9001	101	Jazz Time	555-9928	.....
9002	100	The Mike Hernandez Trio	959-8837	.....
9003	100	The Country Squires	709-3542	.....

Figure 3.14 *An example of a one-to-many relationship*

Many-to-Many Relationships

Students

Student ID	Student First Name	Student Last Name	Student Phone	*<< other fields >>*
60001	Zachary	Erlich	553-3992	.....
60002	Susan	Black	790-3992	.....
60003	Joe	Rosales	551-4993	.....

Classes

Class ID	Class Name	Instructor ID	*<< other fields >>*
900001	Intro. to Political Science	220087	.....
900002	Adv. Music Theory	220039	.....
900003	American History	220148	.....

Figure 3.15 *An example of an unresolved many-to-many relationship*

Unresolved = how do you ***easily*** *associate a student with all their classes, or a class with all its students?*
Answer: This information can only be recorded in the linking (middle) table, NOT in any of the two tables

Students

Student ID	Student First Name	Student Last Name	Student Phone	*<< other fields >>*
60001	Zachary	Erlich	553-3992	.....
60002	Susan	Black	790-3992	.....
60003	Joe	Rosales	551-4993	.....

Student Schedule (*Linking Table*)

Student ID	Class ID
60003	900001
60001	900003
60003	900003
60002	900002
60001	900001

Classes

Class ID	Class Name	Instructor ID	*<< other fields >>*
900001	Intro. to Political Science	220087	.....
900002	Adv. Music Theory	220039	.....
900003	American History	220148	.....

Figure 3.16 *Resolving the many-to-many relationship with a linking table*

2. Mandatory/Optional Participation

-Ex. Make Agent table as mandatory participation in the database because agent 100 must exist in Agent table before using it in the Client table

[this can be enforced with a fk_constraint, we will see later]

3. Degree of Participation

-Ex. CLIENTS 1,8 = one agent must manage at least one client but no more than 8

-Ex. AGENTS 1,1 = a client must be assigned to only one agent

Agents

Agent ID	Agent First Name	Agent Last Name	Date of Hire	Agent Home Phone
100	Stella	Rosales	05/16/09	553-3992
101	Steve	Horst	10/15/09	790-3992
102	Samantha	Nathanson	03/01/11	551-4993

Clients

Client ID	Agent ID	Client First Name	Client Last Name	Client Home Phone
9001	100	Stewart	Jameson	553-3992
9002	101	Susan	Black	790-3992
9003	102	Scott	Baker	551-4993

Figure 3.17 *The AGENTS and CLIENTS tables*

D. Integrity-Related Terms

1. Field specifications
2. Data integrity – ex. no duplicate records, consistent entries like Dr. vs Doctor

Employees

Employee ID	Employee First Name	Employee Last Name	Home Phone	*<< other fields >>*
100	Zachary	Erlich	553-3992	.....
101	Susan	Black	790-3992	.....
102	Joe	Rosales	551-4993	.....

Compensation

Employee ID	Hourly Rate	Commission Rate	*<< other fields >>*
100	19.75	3.5%	.....
101	25.00	5.0%	.....
102	22.50	5.0%	.....

Figure 3.13 *An example of a one-to-one relationship*